

Criminal Dataset Analysis

1. Problem Statement

The objective of this analysis is to identify patterns, trends, and risk indicators within criminal encounter records using demographic, behavioral, and operational data. The analysis aims to help understand factors such as threat levels, age distribution, race, gender, mental illness indicators, and operational response patterns for better decision-making and data-driven insights.

2. Data Dictionary

Column Name	Description
Id	Unique identifier for each incident record.
Name	Name of the individual involved in the incident.
Date	Date on which the incident occurred.
Manner Of Death	Method or type of death reported in the incident.
Armed	Indicates the type of weapon or object carried by the individual.
Age	Age of the individual involved in the incident.
Gender	Gender of the individual.
Race	Ethnicity or racial background of the individual.
City	City where the incident took place.
State	State where the incident occurred.
Signs Of Mental Illness	Indicates whether signs of mental illness were observed.
Threat Level	Threat level assigned during the incident.
Flee	Status showing whether the individual attempted to flee.
Body Camera	Shows whether body camera evidence was available during the incident.

3. Important KPIs / Measures

- **Total Criminal Incident Records**

Measures the total number of incidents available in the dataset for overall monitoring.

- **Dominant Age Group Involved in Incidents**

Represents the age group most frequently involved in criminal incidents.

- **Incidents Involving Mental Illness**

Tracks cases where signs of mental illness were observed to analyse behavioral patterns.

- **Mental Illness Ratio**

Compares the frequency of armed and unarmed individuals involved in incidents.

- **Most Affected State**

Identifies the state with the highest number of criminal incidents.

4. Storytelling Visualizations

1. **Crimes Distribution Over Armed Type**

Shows how criminal incidents are distributed based on different armed categories or weapon types.

2. **Top 5 Crime Affected States**

Highlights the five states with the highest number of criminal incidents.

3. **Crimes Distribution by Age Group & Race**

Represents the distribution of criminal incidents across different age categories & races.

4. **Analyze overall Data Gender**

Allows dynamic analysis of criminal incidents based on different gender categories using interactive filtering.

5. **Analyze overall Data Gender**

Displays the geographical distribution of criminal incidents across different states using map-based visualization.